Code No: 158AK **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
COMPUTATIONAL COMPLEXITY
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Differentiate between Class P and Class NP.
b) What do you mean by polynomial run time algorithm? Explain. [7+8]

2.a) Explain the Reducibility with example.
b) How do you calculate computational complexity? Discuss. [8+7]

3.a) Write the Quick sort algorithm? Analyze the time complexity in worst case.
b) Explain about travelling sales person problem. Explain how it can be solved using Branch and Bound method. Show the steps and solution by taking any one example containing at least 5 cities and 10 ways to travel between them. [5+10]

4.a) Explain matrix chain multiplication using an example.
b) Explain Huffman coding with an example. [8+7]

5.a) How does digital fingerprinting algorithm work? Explain.
b) List and explain the randomized algorithms with example. [8+7]

6.a) Discuss the advantages of randomized algorithm.
b) Elaborate on De-Randomization Advanced Algorithms. [8+7]

7.a) Write a brief note on approximation algorithm.
b) What is a Spanning tree? What are its applications? Explain. [8+7]

8.a) How elements are inserted and deleted in B-Trees? Explain.
b) Construct an AVL tree with the following numbers: 25, 46, 13, 55, 15, 30, 58, 4, 6. Insert 50, 10 and 40, delete 25, 13 and 30 and rebalance the tree if necessary in each case. [8+7]

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